Vitali Shpakowski

AI-Driven Tech Lead | Senior Software Engineer

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Location: Minsk, Belarus, UTC+3 (Open to relocation)
Languages: Belarusian/Russian (Native), English (B1+/B2)

PROFILE

Technical leader with 15 years of experience in designing and launching high-scale systems.

Expert in microservices and micro-frontend architectures powered by Node.js, React, and Angular. Focused on delivering business value through rigorous architectural oversight, ensuring code integrity and security in AI-assisted environments. Extensive expertise in data management and deploying resilient cloud infrastructure with full CI/CD automation.

Specialized in AI-Driven Development (AIDD): managing the full product lifecycle via Antigravity / Cursor orchestration and custom agentic workflows based on Anthropic patterns.

Looking for a challenging role as an AI-Driven Tech Lead or Solutions Architect where I can leverage my expertise to build next-gen AI-integrated products.

SKILLS

AI & LLM: AI-Driven Development (AIDD), AI Orchestration, Prompt Engineering, RAG (Retrieval-Augmented Generation), Antigravity / Cursor, Anthropic patterns, Gemini CLI, AI-agents workflow optimization, AI-assisted Code Review & Final Solution Validation.

Architecture & Leadership: System Design, Solutions Architecture (Microservices, Microfrontends), API Design, Technical Roadmap, Team Leadership (2–5 devs), Stakeholder Communication, Technical Grooming.

Frontend: TypeScript, Angular + RxJS, Next.js & React (SSR/SSG), State Management (NgRx, Zustand, TanStack Query), Tailwind CSS, SASS/SCSS, Module Federation (Microfrontends), Web Vitals & Performance Optimization.

Backend & Databases: Node.js, NestJS, Express.js, PostgreSQL, TypeORM, Firebase.

Cloud & DevOps: Google Cloud Platform (GCP), Oracle Cloud, AWS, Kubernetes, Docker, Terraform, Nginx, GitHub Actions (CI/CD), SonarQube.

Quality & Testing: Unit/Integration/E2E Testing (Jest, Jasmine, Karma), Architectural Supervision, Automated Quality Gates.

WORK EXPERIENCE

AI-Driven Tech Lead

June 2025 — Present

MCP COOP DAO

Project Role: Tech Lead & Full-Stack Developer

Designed and engineered a scalable decentralized ecosystem — delivering 3 independent production applications — integrating a DAO platform, Blockchain protocols, Telegram Mini Apps, and a React Native mobile app.

Led a lean, high-performing team of 3 specialists. Accelerated full-stack development and drastically reduced Time-to-Market by integrating AI orchestration (Antigravity/Cursor, custom AI workflows) into the engineering pipeline.

- **AI-First Engineering:** Delivered 90%+ of the entire system logic using custom agentic workflows, achieving unprecedented development density and code consistency.
- **Blockchain & Backend:** Engineered the platform core in Go (Golang) for blockchain operations and a high-performance backend using NestJS + Prisma + PostgreSQL. Integrated Open API and Firebase for hybrid cloud logic.
- **Multi-Platform Architecture:** Designed a reusable micro-frontend system that powers independent web applications, Telegram Mini Apps (TMA SDK), and mobile interfaces built with React Native.
- **Identity & Security:** Implemented a complex identity layer (SSO, Role-based Access, JWT) featuring seamless integration between Telegram accounts and standard web authentication protocols.
- **Multi-cloud CI/CD:** Orchestrated full delivery pipelines using GitHub Actions and Docker containerization. Deployed a resilient infrastructure across a hybrid cloud environment (Google Cloud Platform and Oracle Cloud) for maximum availability.

Stack: Go, Next.js (Turbopack), NestJS, React Native, Telegram SDK, Prisma, PostgreSQL, Redis, Firebase, Cloudflare, Docker, GitHub Actions, GCP, Oracle Cloud.

Senior Software Engineer

December 2013 — December 2024

EPAM Systems

Minsk, Belarus (2013–2021) | Tbilisi, Georgia (2021–2024)

SAP Coveo AI Search Platform

Project Role: Team Lead & Full-stack Developer

Led a cross-functional team of 4 in developing an enterprise AI-powered search platform. Designed and implemented a BFF (Backend for Frontend) architecture to serve as a high-performance middleware between the UI and external AI services.

- **Resilient AI Orchestration:** Engineered a feature-flag system with a custom fallback mechanism on the backend, enabling seamless switching and high availability between OpenSearch and Coveo AI services.
- **BFF & Caching:** Developed a NestJS middleware layer, integrating Redis caching to optimize search query performance and reduce external API latency.
- **GenAI Integration:** Integrated a generative AI response system into the search interface, ensuring robust delivery of dynamic, AI-generated content.
- **Leadership & Management:** Performed over 50% of BA/PM duties, including stakeholder management, backlog prioritization, and comprehensive technical documentation.

Stack: Angular, NestJS, Redis, TypeScript, RxJS, CoveoUI, OpenSearch.

US Bank Loyalty Program (MVP)

Project Role: Full-stack Developer (Key Developer)

Engineered an end-to-end loyalty platform MVP for integration with US banking systems via Open API, managing the full lifecycle from database design to CI/CD.

- **Hybrid Next.js Architecture:** Implemented SSR for SEO-critical promotional pages and a React-based client-side dashboard for administrative management.
- **Security & Auth:** Developed a secure identity layer using OAuth 2.0 (Google), OTP, and JWT with granular role-based access control.
- **Database Management:** Built a flexible data schema using Prisma ORM and PostgreSQL to handle complex promotional logic and user rewards.
- **DevOps:** Automated deployment workflows via Docker and GitHub Actions, ensuring a rapid and reliable delivery cycle.

Stack: Next.js, React, Node.js, Prisma, PostgreSQL, OAuth 2.0, Docker, GitHub Actions.

Carrefour — E-commerce Platform

Project Role: Key Developer & Technical Coordinator

Led the development of critical features for a high-traffic e-commerce store (500K+ monthly users) during a strategic platform transition. Combined high-level technical execution with project management responsibilities.

- **Product Impact:** Engineered high-conversion features, including advanced product filtering and an optimized checkout flow, directly improving user experience and sales metrics.
- **DevOps & Quality:** Established a CI/CD pipeline integrated with Jenkins and SonarQube, significantly reducing code quality issues and automating delivery.
- **Stakeholder Management:** Acted as a technical bridge between business analysts and the dev team, managing backlog prioritization, technical grooming, and requirement analysis.
- **Team Coordination:** Coordinated and held full responsibility for a sub-team of 2 developers, ensuring rigorous code standards and consistent delivery within an Agile/Scrum framework.

Stack: JavaScript (ES6+), jQuery, SASS, Jenkins, SonarQube, Git, Agile.

Refinitiv — Financial Trading Platform

Project Role: Key Developer

Delivered two Angular-based web applications, including a major migration of a legacy C# desktop solution to the web. Coordinated technical tasks for a sub-team of 3 developers, focusing on architecture and data integration.

- **API Transformation:** Engineered a custom transformation layer using Express.js to bridge legacy SOAP services with a modern REST API, enabling seamless data migration from desktop to web.
- **Real-time Services:** Designed the frontend architecture to handle high-frequency, real-time data streams via WebSocket connections.
- **Performance:** Achieved a 60% reduction in initial load time (from 5s to 2s) by implementing lazy loading and advanced code-splitting strategies.
- **Testing & Quality:** Maintained high engineering standards with 80%+ code coverage using Jasmine and Karma, ensuring enterprise-grade stability.

Stack: Angular, TypeScript, Express.js, RxJS, WebSockets, SOAP, REST API, Jasmine, Karma.

MoneySuperMarket — Financial Services Aggregator

Project Role: Key Developer & Supervisor

Acted as a Key Developer for a high-traffic platform (10M+ users/mo) and Supervisor for a dedicated UI sub-team of 4. Engineered the core system using AngularJS and successfully orchestrated a strategic migration to Angular 2+ to enhance scalability.

- **Architecture:** Designed a micro-frontend system and its seamless integration into the Adobe CQ5 (AEM) enterprise platform, reducing deployment time by 40%.
- **Performance:** Optimized initial load time from 6.5s to 3.8s (42% improvement) through advanced asset tuning and performance optimization.
- **Quality Control:** Developed a library of 50+ reusable components using BEM and established code review standards for the UI team, decreasing production bugs by 60%.
- **Team Supervision:** Facilitated technical grooming and sprint planning for the UI group, maintaining a 95% sprint delivery rate in an Agile/Scrum environment.

Stack: Angular 2+, AngularJS, TypeScript, RxJS, Redux, NgRx, BEM, SASS, Adobe CQ5, Jenkins.

Middle Frontend Developer

April 2013 — December 2013

Startup Labs

Developed high-performance responsive promotional sites for financial services. Implemented complex web forms with robust client-side validation, improving data accuracy. Engineered mobile-first layouts using SASS, ensuring full cross-browser compatibility and optimized performance.

Stack: JavaScript (jQuery), HTML5/CSS3, SASS, WordPress, Twig, Git.

Junior Frontend Developer

April 2012 — March 2013

Axamit Versa

Developed responsive interfaces and interactive UI components using Adobe AEM (CQ5). Built a reusable component library, significantly reducing development time for common UI patterns across client projects.

Stack: JavaScript (jQuery), Adobe AEM, HTML5/CSS3, Git.

CERTIFICATIONS

EPAM Systems

Advanced Software Engineering: Software Development Processes (SDLC), Backend Development (Node.js, Express.js, MongoDB) / UI Architecture Series: Enterprise-level Engineering Excellence (Angular stack core).

EDUCATION

Systems Engineer

Belarusian State University of Informatics and Radioelectronics (BSUIR) · 2006–2011 · Minsk, Belarus